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VEHICLE DISPLAY SYSTEM

Abstract

Provided is a vehicle display system including a projector configured to emit projection light to a display mirror configured to reflect the projection light and transmit first reflected light to an optical reflector, as second reflected light reflected on the optical reflector is incident, to display a virtual image that is visible to a user in a vehicle; and the optical reflector positioned in one region in the vehicle to reflect the first reflected light to the display mirror. A polarizing filter having a polarization axis in a first direction is coated on an external lens emitting the projection light. The display mirror is one region of a windshield, made of glass, a window, and a transparent flat plate, made of glass or acrylic, and a polarizing filter having a polarization axis in a second direction perpendicular to the first direction is coated on an inside of the display mirror.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0025472, filed Feb. 22, 2024, Korean Patent Application No. 10-2024-0059449, filed on May 3, 2024, and U.S. Patent Application No. 63/568,636 dated Mar. 22, 2024, the entire disclosures of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to a display system installed inside a vehicle, and more particularly to a technology for providing various display images using a windshield, etc. in a vehicle.

2. Description of the Related Art

[0003] Displays have been installed and provided in vehicles to assist the driver's driving or offer displays for entertainment.

[0004] In particular, as autonomous driving technology has been intensively studied recently, the use of displays for various purposes other than driving in vehicles has increased. For this reason, display screens installed in vehicles are gradually becoming larger.

[0005] While an increasingly enlarged display screen has the advantage of improving the user's image visibility, there are disadvantages in that the display screen occupies a very large area inside a vehicle, which reduces the driver's driving concentration or obstructs the airflow of a vehicle's air ventilation system.

[0006] In addition, HeadUpDisplays (HUDs) are also widely used to display essential information necessary for driving without lowering the driver's concentration in a vehicle. However, HUDs have a very small screen, making them unsuitable for continuous video viewing, having a very narrow viewing angle, and using a high-power, high-brightness light source to maintain visibility even during the day.

[0007] Meanwhile, optical modulation devices for changing the light transmission, reflection, and scattering characteristics, or for modulating the phase, amplitude, polarization, intensity, and path, are being utilized for various purposes. As such optical modulation devices, microelectromechanical system (MEMS) structures that utilize the mechanical micro-motion of light blocking and reflection are being studied. In particular, structures such as metagrating or metasurfaces that induce optical modulation by utilizing the surface plasmon resonance phenomenon for incident light have been studied in recent years.

RELATED ART DOCUMENT

Non-Patent Document

[0008] Metagratings: Beyond the Limits of Graded Metasurfaces for Wave Front Control, Younes Ra'di, Dimitrios L. Sounas, and Andrea Alu, Phys. Rev. Lett. 119, 067404-Published 10 Aug. 2017

SUMMARY OF THE INVENTION

[0009] Therefore, the present disclosure has been made in view of the above problems, and it is an

object of the present disclosure to provide a vehicle display system that occupies a minimum space in a vehicle and uses glass installed in the vehicle to create a wide screen.

[0010] In accordance with the present disclosure, the above and other objects can be accomplished by the provision of a vehicle display system.

[0011] The vehicle display system includes a projector configured to emit projection light toward an optical reflector; an optical reflector positioned in one region in a vehicle to reflect the projection light; and a display mirror configured to display a virtual image, which is visible to a user, in a vehicle as the light reflected on the optical reflector is incident.

[0012] In the projector, a polarizing filter having a polarization axis in a first direction is coated on an external lens emitting the projection light.

[0013] The display mirror is one region of a windshield, made of glass, a window, and a transparent flat plate, made of glass or acrylic, in the vehicle.

[0014] A polarizing filter having a polarization axis in a second direction perpendicular to the first direction is coated on an inside of the display mirror.

[0015] The projector is positioned in one region of an overhead console located at a top of a vehicle based on a driver, pillar A in the vehicle, pillar B in the vehicle, an armrest of a rear row of the vehicle, and a ceiling of the rear row of the vehicle.

[0016] The optical reflector is positioned in one region of a dashboard and ceiling in the vehicle and an inside of a door.

[0017] The display mirror is one region of the windshield.

[0018] The optical reflector includes a meta-surface on which a plurality of nanostructures are arranged and which diffracts and reflects the projection light according to a shape and arrangement form of each of the arranged nanostructures.

[0019] The nanostructures have a thickness thinner than a wavelength of the projection light.

[0020] An arrangement distance between the nanostructures is shorter than the wavelength of the projection light.

[0021] The optical reflector includes: a substrate; and a plurality of nanostructures arranged on the substrate and configured to function as a meta-surface.

[0022] The substrate includes a transparent film layer configured to transmit the projection light; a reflective layer positioned under the transparent film layer; and an adhesive layer positioned under the reflective layer.

[0023] The optical reflector further includes a phase synchronization part configured to apply voltage to each of the nanostructures in conjunction with a controller of the projector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The above and other objects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0025] FIG. 1 illustrates the conceptual diagram of a vehicle display system according to one embodiment;

[0026] FIG. 2 illustrates the side view of an optical reflector according to an embodiment;

[0027] FIG. 3 illustrates the lamination drawing of a substrate of the optical reflector according to FIG. 2;

[0028] FIG. 4 illustrates the plan view of the optical reflector according to an embodiment;

[0029] FIG. 5 is a block diagram illustrating functional parts of a projector according to an embodiment;

[0030] FIG. 6 illustrates another installation example of the vehicle display system according to

one embodiment.

[0031] FIG. 7 is a conceptual diagram for explaining the structure of an optical reflection-type small projector according to an embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0032] Since the present disclosure may be applied with various modifications and may have various embodiments, exemplary embodiments and drawings of the present disclosure are intended to be explained and exemplified. However, these exemplary embodiments and drawings are not intended to limit the embodiments of the present disclosure to particular modes of practice, and all changes, equivalents, and substitutes that do not depart from the spirit and technical scope of the present disclosure should be understood as being encompassed in the present disclosure. Like reference numerals refer to like elements in describing each drawing.

[0033] The terms such as “first,” “second,” “A” and “B” are used herein merely to describe a variety of constituent elements, but the constituent elements are not limited by the terms. The terms are used only for the purpose of distinguishing one constituent element from another constituent element. For example, a first element may be termed a second element and a second element may be termed a first element without departing from the teachings of the present disclosure. The term “and/or” includes any or all combinations of one or more of the associated listed items.

[0034] It should be understood that when an element is referred to as being “connected to” or “coupled to” another element, the element may be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected to” or “directly coupled to” another element, there are no intervening elements present.

[0035] The terms used in the present specification are used to explain a specific exemplary embodiment and not to limit the present inventive concept. Thus, the expression of singularity in the present specification includes the expression of plurality unless clearly specified otherwise in context. Also, terms such as “include” or “comprise” should be construed as denoting that a certain characteristic, number, step, operation, constituent element, component or a combination thereof exists and not as excluding the existence of or a possibility of an addition of one or more other characteristics, numbers, steps, operations, constituent elements, components or combinations thereof.

[0036] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0037] Hereinafter, the present disclosure will be described in detail by explaining exemplary embodiments of the disclosure with reference to the attached drawings.

[0038] FIG. 1 illustrates the conceptual diagram of a vehicle display system according to one embodiment.

[0039] Referring to FIG. 1, a vehicle display system **100** according to one embodiment may include a projector **10** configured to emit projection light toward a region placed in front of a vehicle, an optical reflector **20** placed in the region to reflect the projection light emitted from the projector **10**, and a display mirror DM configured to display a virtual image that can be seen by a user in the vehicle when reflected light reflected from the optical reflector **20** is incident.

[0040] In an embodiment, the projector **10** may be a Digital Light Processing (DLP)-type projector. The DLP-type projector can express the brightness of each pixel by controlling the time for which light is reflected through tilt transition corresponding to the on/off state of micro-mirrors using a Digital Micromirror Device (DMD) chip corresponding to a semiconductor optical switch in which micro-actuated mirrors are integrated.

[0041] In another embodiment, the projector **10** may be a Laser Beam Scanning (LBS)-type projector. The LBS-type projector simultaneously scans the projection pixel brightness of laser used as a light source, projects it, and reflects it, so it provides relatively higher ANSI performance than the DLP-type projector that requires a large light source with high power consumption, and has the advantage of a high contrast ratio.

[0042] Here, in the projector **10**, a polarizing filter having a polarization axis in a first direction may be coated on a part (e.g., an external lens) where projection light is emitted. The polarizing filter having the polarization axis in the first direction may be a film that allows light polarized in the first direction to pass through and blocks the remaining light. For example, the polarizing filter having the polarization axis in the first direction may transmit S-polarized light.

[0043] In an embodiment, the projector **10** may be positioned adjacent to a fixed installation portion of a rearview mirror in a vehicle. For example, the projector **10** may be positioned at the front or rear end of the fixed installation portion of the rearview mirror so as to face a dashboard.

[0044] In another embodiment, the projector **10** may be positioned so as to face the dashboard from an overhead console located at the top relative to a driver in the vehicle. For example, the overhead console may be interpreted as including the ceiling area, where a sun visor blocking the sunlight is installed, of the upper part of the vehicle; and the fixed installation portion of the rearview mirror.

[0045] In another embodiment, the projector **10** may be positioned on pillar A in the vehicle so as to face the dashboard.

[0046] In another embodiment, the projector **10** may be positioned on pillar B in the vehicle so as to face the dashboard. Specifically, the projector **10** may be positioned on an upper side of the pillar B in the vehicle.

[0047] That is, the projector **10** may be positioned at the rear of the vehicle with respect to the dashboard DB, and then emit projection light toward the optical reflector **20** (or the dashboard DB where the optical reflector **20** is positioned).

[0048] In addition, the projector **10** may emit projection light, instead of the optical reflector **20**, toward the display mirror DM.

[0049] The optical reflector **20** may reflect the projection light, emitted from the projector **10**, toward the display mirror DM. In still another embodiment, the optical reflector **20** may receive the reflected light from the display mirror DM and reflect the received reflected light to the display mirror DM. In particular, the optical reflector **20** may include a meta-surface where a plurality of nanostructures are arranged. The meta-surface diffracts and reflects projection light depending upon the shape and arrangement shape of each of the arranged nanostructures.

[0050] The optical reflector **20** may be positioned in one area of a dashboard and ceiling inside a vehicle and the inside of a door.

[0051] The meta-surface, which is a structure with a very small nano-unit size, is a two-dimensional form of metamaterial that is a general term for structures that induce light wave phenomena that are impossible with natural materials. Nanostructures of extremely small thickness are arranged, and depending on the shape and arrangement of the arranged nanostructures, the properties of light, such as amplitude, wavenumber, phase, and polarization, can be changed very rapidly.

[0052] The display mirror DM may be one region of a windshield, made of glass, a window, and a transparent flat plate, made of glass or acrylic, within a vehicle.

[0053] That is, the display mirror DM may not be an independent device placed in a vehicle, but may be one region of a windshield, made of glass, a window, and a transparent flat plate, made of glass or acrylic in the vehicle. For example, it may be one region of a windshield. That is, a virtual image can be recognized by a user through a windshield functioning as the display mirror DM by reflected light incident on one region of the windshield.

[0054] A polarizing filter having a polarization axis in a second direction may be coated on the inside of one region (in the inward direction of the vehicle) of the display mirror DM. The second

direction may be perpendicular to the first direction, and the polarization filter having a polarization axis in the second direction may allow P-polarized light to pass through.

[0055] That is, since the polarizing filter having a polarization axis in the second direction perpendicular to the first direction is coated on the inside of the display mirror DM, a display image displayed through the display mirror DM from the outside of the vehicle may not be identified.

[0056] The display mirror DM may receive the reflected light from the optical reflector **20**, and display the received reflected light to a user in the form of a virtual image.

[0057] In still another embodiment, the display mirror DM may receive the projection light from the projector **10**, and reflect the received projection light to transmit it to the optical reflector **20** as reflected light. The display mirror DM is coated with a polarizing filter in a direction perpendicular to the polarizing filter coated on an external lens of the projector **10**, so that the projection light may not pass through the display mirror DM and may be reflected as reflected light toward the optical reflector **20**.

[0058] In addition, the polarizing filter coated on the inside of the display mirror DM may help to minimize the influence of sunlight incident from the outside to the inside of the vehicle, thereby providing a clear display image.

[0059] That is, according to the first example of the present disclosure, the projector **10** may emit projection light to the display mirror DM, the display mirror DM may reflect the projection light as reflected light to the optical reflector **20**, and the optical reflector **20** may reflect the received reflected light to the display mirror DM. In this case, since the path of light travels longer, there is an advantage that a large screen can be displayed within a minimum space.

[0060] In addition, according to the second example of the present disclosure, the projector may emit projection light to the optical reflector **20**, and the optical reflector **20** may reflect the projection light as reflected light toward the display mirror DM. In this case, although the light travel path is relatively shorter than in the above-described example, the relative positions of the projector **10**, the optical reflector **20**, and the display mirror DM are configured such that the reflected light is transmitted only from the optical reflector **20** to the display mirror DM, so that the convenience of construction is relatively high.

[0061] Hereinafter, a description is provided based on the first example that the optical reflector **20** receives projection light, but the case based on the second example that the optical reflector **20** receives reflected light (hereinafter referred to as the first reflected light) from the display mirror DM and transmits the reflected light (hereinafter referred to as the second reflected light) to the display mirror DM can also be applied.

[0062] FIG. 2 illustrates the side view of an optical reflector according to an embodiment. FIG. 3 illustrates the lamination drawing of a substrate of the optical reflector according to FIG. 2. FIG. 4 illustrates the plan view of the optical reflector according to an embodiment.

[0063] In FIGS. 2 to 4, for convenience of explanation, the front direction of the vehicle is described as the direction in which the display mirror DM is located based on the optical reflector **20** when the optical reflector **20** is placed on the dashboard DB and the display mirror DM is used as one region of a windshield, but it should be interpreted that it can be changed depending upon the installation location described in this specification.

[0064] Referring to FIGS. 2 to 4, the optical reflector **20** may include a substrate SUB; and a plurality of nanostructures NSs arranged on the substrate SUB and configured to function as a meta-surface.

[0065] In an embodiment, the substrate SUB may include a transparent film layer **21** configured to transmit projection light, a reflective layer **22** disposed under the transparent film layer **21**, and an adhesive layer **23** disposed under the reflective layer **22**.

[0066] For example, the transparent film layer **21** may be an optically clear resin layer (OCR).

[0067] In the reflective layer **22**, a plurality of transparent tubes or transparent balls based on silicon dioxide (SiO₂) may be positioned at predetermined distribution intervals (e.g., at

predetermined left-right intervals, etc.) on the reflective layer **22**. Here, a reflective coating layer may be arranged on the inner lower curvature regions (refers to the lower curved regions of the balls or tubes) of the transparent tubes or balls of the reflective layer **22**.

[0068] For example, the reflective coating layer may be a mirror coating layer based on a metallic material, etc. Therefore, when the projection light passes through the transparent film layer **21** and reaches the reflective layer **22**, it is reflected through the reflective coating layer, and the reflected light can be projected to the outside of the substrate SUB.

[0069] In an embodiment, the adhesive layer **23** may be formed using an optical acrylic adhesive composition.

[0070] A plurality of nanostructures NSs may be arranged along a predetermined array interval d in the longitudinal direction of the vehicle (a direction crossing between the front and rear of the vehicle).

[0071] Here, the plural nanostructures NSs may have a shape dimension of a sub-wavelength smaller than the wavelength of the projection light emitted from the projector **10**. For example, the thickness t of the nanostructures NS may be smaller than the wavelength of projection light. In addition, an arrangement distance d between neighboring nanostructures NS among nanostructures NSs may be smaller than the wavelength of the projection light.

[0072] In particular, the arrangement distance of the nanostructures NS may be $\frac{1}{2}$ (or $\frac{1}{3}$) or less the wavelength of the projection light. By arranging the nanostructures NSs to be $\frac{1}{2}$ or less the wavelength of the projection light, each of the nanostructures NS can function as a basic unit of the meta-structure that changes the properties of the light of the projection light.

[0073] The nanostructures NS may be configured to have a predetermined slope in a predetermined direction. For example, they may be configured to have the predetermined slope toward the windshield. Here, “have the predetermined slope toward the windshield” may mean that the slope is arranged parallel to the surface of the windshield or arranged to have a predetermined acute angle (within 45 degrees) based on the parallel arrangement.

[0074] In the nanostructures NS, the slope may be provided with an absorption layer ABS that absorbs external light. The absorbing layer may be surface-coated with a known optical coating material that absorbs light, or the absorbing layer material itself may be made of a conventional light-absorbing material.

[0075] In an embodiment, the nanostructures NS may be configured as rods extending in the width direction (i.e., left and right directions of a specific seat) of the vehicle, and may be shaped like a rod cut to have the slope. In this case, the thickness t of the nanostructure NS arranged in the center among the nanostructures NSs may be the thickest, and the thickness t of the nanostructure NS may gradually increase as it moves away from the center. For example, it can be configured to increase by a multiple of 1 or more every time the predetermined interval increases from the center.

[0076] In another embodiment, the nanostructures NS may be formed as cylinders extending in the direction of the vehicle width, and may have a shape that is cut from the cylinder to have the slope.

[0077] Although not shown in the drawings, the nanostructures NS are formed as cylinders, rod-shaped columns, triangular columns, etc., protruded to have a longitudinal direction perpendicular to the substrate SUB, and may have a shape cut to have the slope. In this case, the nanostructures NS may be configured such that the thickness t is thicker as they get closer to the center of the optical reflector **20**, and the thickness t gradually increases as they get farther from the center. In addition, the spacing between the nanostructures NS may also be configured such that the spacing between the nanostructures NS becomes smaller as it approaches the center of the optical reflector **20** and increases as it moves away from the center of the optical reflector **20**.

[0078] The nanostructures NS may be made of a dielectric material. For example, the nanostructures NS may be any one selected from single crystal silicon, polycrystalline silicon, amorphous Si, Si.sub.3N.sub.4, GaP, TiO.sub.2, AlSb, AlAs, AlGaAs, AlGaInP, BP, and ZnGeP.sub.2 having a refractive index greater than that of the substrate SUB.

[0079] In another embodiment, the nanostructures NS may be an alloy composed of one or more combinations of copper, aluminum, iron, cobalt, zinc, titanium, platinum, silver, gold, palladium, and iridium as conductive materials.

[0080] The absorption layer ABS can minimize the influence of reflected light by external light by absorbing external light coming through the windshield.

[0081] Meanwhile, the optical reflector **20** may further include a phase synchronization part (not shown) that applies voltage to each of the nanostructures NS in conjunction with a controller of the projector **10**.

[0082] The phase synchronization part may induce an optical phase change of the nanostructures NS by applying a voltage that changes according to a cycle (e.g., a cycle identical to the image display cycle of the projector **10**, a cycle obtained by multiplied the image display cycle by a predetermined proportional constant defined in advance, etc.) corresponding to the operation of the projector **10** to each of the nanostructures NSs, thereby allowing the nanostructures NS to operate in synchronization with the operation of the projector **10**. The phase synchronization part may be electrically connected to a power line within the vehicle.

[0083] Specifically, the phase synchronization part may classify the plural nanostructures NSs into a plurality of groups, and match each of a plurality of pixels constituting a display screen output from the projector **10** to each of the classified groups. That is, one group corresponding to one pixel may be set. For example, phase synchronization operates such that, for each cycle corresponding to the image display cycle of the projector **10**, a voltage (where the voltage corresponding to the color is predefined such that the nanostructures of the corresponding group have a specific reflection steering angle for the corresponding pixel) corresponding to the color of the pixel matched with the i-th group (where i is a natural number from 1 to n, n is the total number of groups) is applied to the i-th group among the plural groups.

[0084] When the phase synchronization part performs phase control of the nanostructures NS by applying voltage according to the display screen of the projector **10**, the reflected light that is reflected by the projection light incident on the substrate SUB may be reflected with the emission angle steered according to the reflection steering angle. Here, the steered reflected light may be referred to as steered light.

[0085] In an embodiment, through the control of the nanostructures by the phase synchronization part, the incident angle of the projection light based on the surface normal vector (a line perpendicular to the surface of the substrate) of the substrate SUB and the emission angle of the steered reflected light, the steered light, may be defined according to Mathematical Equation 1 below:

[00001] $\sin(\theta_{sl}) = \sin(\theta_{il}) \pm \frac{\Delta\theta}{T}$ [Mathematical Equation 1]

[0086] In Mathematical Equation 1, ϵ is the wavelength of the projection light, T is a cycle corresponding to the image display cycle of the projector **10**, $\theta_{sub.il}$ is the incident angle of the projection light, and $\theta_{sub.sl}$ is the emission angle of the steered light.

[0087] Here, the steered light controlled by the phase synchronization part may be diffracted light formed by diffracting the projection light more than once.

[0088] FIG. 5 is a block diagram illustrating functional parts of a projector according to an embodiment.

[0089] Referring to FIG. 5, the projector **10** may include a power supply part **11** electrically connected to a power source installed in a vehicle to provide driving power for the projector **10**, a light source generation part **12** configured to generate a light source, which corresponds to display image data stored inside, by power supplied from the power supply part **11** and emit it as projection light, a vehicle information receiving part **13** configured to obtain vehicle information and multimedia information to be provided as a display screen by being linked with an ECU, an infotainment system, etc. in the vehicle, a display mirror monitoring part **14** configured to monitor

the display screen displayed on the display mirror DM, and a distortion correction part **15** configured to generate distortion correction information according to the monitored display screen. [0090] The vehicle information may be speed, tire pressure, internal temperature, engine RPM, etc., which are essential for vehicle control within the vehicle.

[0091] The multimedia information may refer to radio information, navigation information, etc. provided through an infotainment system within a vehicle.

[0092] The light source generated by the light source generation part **12** may be a laser light source.

[0093] The vehicle information receiving part **13** may obtain vehicle information and multimedia information by communicating with the vehicle via Bluetooth communication, Ethernet communication, CAN communication, etc.

[0094] The display mirror monitoring part **14** may be a camera positioned to capture a display screen within the vehicle, and it may be desirable that the camera is positioned as close to the driver's line of sight as possible.

[0095] In addition, the display mirror monitoring part **14** may be a camera positioned to capture not only the display screen but also the shape of the optical reflector **20** within the vehicle. Through this, the display mirror monitoring part **14** may obtain the curvature or shape of the optical reflector **20**.

[0096] The distortion correction part **15** may generate distortion correction information through the in-vehicle display screen acquired by the display mirror monitoring part **14**. For example, the position and shape of the display screen in the vehicle may be compared with a predefined reference position and reference shape (such as a rectangle) to obtain a difference value, and distortion correction information may be generated to offset the obtained difference value. For example, the distortion correction information may be data for adjusting the position and shape of the display image data in a direction that offsets the difference value.

[0097] The distortion correction part **15** may obtain a difference value by comparing the curvature or shape of the optical reflector **20** with a predefined reference curvature or reference shape, and generate auxiliary distortion correction information to offset the obtained difference value. The auxiliary distortion correction information may be information for correcting pixel values at specific locations constituting the display image data so as to correct the difference in optical reflection caused by differences in the curvature or shape of the optical reflector **20** (i.e., in the case of a dashboard, the curvature of the dashboard may be different for each vehicle).

[0098] The light source generation part **12** may correct the display image data using distortion correction information and/or auxiliary distortion correction information, and generate a light source based on the corrected display image data to emit projection light.

[0099] That is, since the display image data is corrected in real-time in consideration of the visibility and image distortion of a virtual image according to the installation location and curvature of the light source reflector **20** according to the present disclosure, the curvature and material of the windshield, etc. in real-time, the image can be corrected in real-time to provide a clear and accurate display screen.

[0100] FIG. **6** illustrates another installation example of the vehicle display system according to one embodiment.

[0101] Referring to FIG. **6**, a vehicle display system **2000** provided in the rear row (row after the first row) of a vehicle is described.

[0102] The vehicle display system **2000** provided in the rear row (row after the first row) of a vehicle according to FIG. **6** may include a projector **10** configured to emit projection light, an optical reflector **20** positioned in one region of the vehicle to reflect the projection light emitted from the projector **10**, and a display mirror DM configured to display a virtual image that is visible to a user in the vehicle as the reflected light reflected on the optical reflector **20** is incident. In the vehicle display system **2000** according to another installation example, the projector **10** may be disposed on the ceiling of the vehicle's rear row (2nd or 3rd row), the C pillar of the vehicle, an

armrest, etc., and may emit projection light to the optical reflector **20** or the display mirror DM.

[0103] As in the second example described above with reference to FIG. **1**, the vehicle display system **2000** may include a projector **10** configured to emit projection light to a display mirror DM, a display mirror DM configured to reflect projection light and transmit the first reflected light to an optical reflector **20**, and to display a virtual image that is visible to a user in a vehicle as the second reflected light reflected on the optical reflector **20** is incident, and an optical reflector **20** arranged in one region within the vehicle to reflect the first reflected light to the display mirror DM.

[0104] The optical reflector **20** may be positioned in one region of the ceiling of the rear row (two or more rows) of the vehicle.

[0105] The display mirror DM may be arranged in one region of a vehicle seat and may be made of a transparent flat material such as a glass material or an acrylic material. That is, in the embodiment according to FIG. **1**, a glass material already existing in a vehicle is used as the display mirror DM, but in the embodiment according to FIG. **6**, a separate transparent flat material may be constructed as the display mirror DM. However, it is not necessarily limited to this, and the door glass installed on the left or right side of the rear row of the vehicle may be utilized as the display mirror DM.

[0106] The region of the vehicle seat may mean one region of the rear of the seat located in the front row (right in front of the rear row) of the rear row of the vehicle, and, preferably, is arranged close to the eye level of a user.

[0107] The display mirror DM may be installed to be inclined at a predetermined angle toward a user seated in the rear row (e.g., such that the upper edge of the display mirror DM is inclined toward the user) such that it is easy to receive the first reflected light from the optical reflector and receive the second reflected light from the optical reflector **20**.

[0108] FIG. **7** is a conceptual diagram for explaining the structure of an optical reflection-type small projector according to an embodiment.

[0109] A projector **10** according to another embodiment of the present disclosure may be an optical reflection-type small projector with an optical reflector **20** built inside.

[0110] Specifically, referring to FIG. **7**, the projector **10** according to another embodiment may be configured such that the above-mentioned optical reflector **20** is built thereinside, and projection light emitted from a light source generation part **12** is transmitted to the optical reflector **20** in the projector **10**.

[0111] In this embodiment, the optical reflector **20** may be named as a metalens, and the projector **10** may be referred to as a metalens optical system-based projector **10**. The metalens optical system-based projector **10** may be miniaturized by including a metalens thereinside and may project a large image and perform optical system miniaturization.

[0112] In the embodiment according to FIG. **7**, the optical reflector **20** may be positioned in one region of one side inside the projector **10**. For example, the optical reflector **20** may be positioned in one region of the upper side inside the projector **10**.

[0113] In this case, the light source generation part **12** is configured to emit projection light to the optical reflector **20** positioned in one region of one side inside the projector **10**.

[0114] The light source generation part **12** may include a cavity region **123** forming laser cavity and an upper clad layer **121** and lower clad layer **122** respectively arranged above and below the cavity region **123** and spaced apart from each other.

[0115] The cavity region **123** may generate light having a predetermined wavelength band when electrons and holes are injected through the electrode layer. For example, the cavity region **123** may be a group III-V compound semiconductor, a quantum dot based on the corresponding group III-V chemical semiconductor, etc.

[0116] Although an electrode layer is not shown in the drawings, it is composed of a first electrode layer and second electrode layer, which are respectively arranged on the upper side of the upper clad layer **121** and the lower side of the lower clad layer **122**, for current injection into the cavity region **123**. So current injection into the cavity region **123** can be performed through a voltage

applied between the first electrode layer and the second electrode layer.

[0117] Meanwhile, the cavity region **123** may include a first reflection region **123** and second reflection region **125** spaced apart from each other with a central cavity **126** as the center, in a direction (i.e., the emission direction of projection light in FIG. 7) crossing the cavity region **123** between the upper clad layer **121** and the lower clad layer **122**.

[0118] On the upper and lower surfaces of the first reflection region **123** and second reflection region **125**, the plural nanostructures NSs described above may be arranged according to the predetermined array interval and thickness described above, etc., and are configured to function as meta-surfaces in the same manner, but have a predetermined reflectivity.

[0119] For example, the first reflection region **123** and the second reflection region **125** may be configured to have the same reflectivity. More preferably, the first reflection region **123** may be configured to have a reflectivity higher than a predetermined reference value, and the second reflection region **125** may be configured to have a reflectivity lower than that of the first reflection region **123**.

[0120] In an embodiment, the light source generation part **12** may be a laser diode.

[0121] The optical reflector **20** reflects the projection light emitted from the light source generation part **12** and transmits the reflected light to the outside through the external lens, and the arrangement direction of the projector **10** may be adjusted such that the reflected light reaches the display mirror DM.

[0122] In the case of the projector **10** according to FIG. 7, since the optical reflector **20** is placed inside the projector **10**, the installation difficulty can be reduced, and a sufficiently large, high-definition screen can be created inside the projector **10** through the optical reflector **20**.

[0123] As apparent above, since a miniaturized projector is used when using a vehicle display system according to the present disclosure, the area occupied by a display system within a vehicle can be minimized.

[0124] In addition, since it is possible to enlarge a screen and provide maximum clarity by using an optical reflector with a minimized area based on a meta-structure, there are advantages in that it is possible to provide a much larger screen than existing HUDs and it can be used for entertainment purposes in addition to essential driving information.

[0125] Further, by using polarization filters and optical patterns, it is possible to provide a display screen with a focused field of view by blocking the possibility of external visibility other than that of an intended user.

[0126] The methods according to the embodiments of the present disclosure may be implemented in the form of a program command that can be executed through various computer means and recorded in a computer-readable medium. The computer-readable medium can store program commands, data files, data structures or combinations thereof. The program commands recorded in the medium may be specially designed and configured for the present disclosure or be known to those skilled in the field of computer software.

[0127] Examples of a computer-readable recording medium may include hardware devices such as ROMs, RAMs and flash memories, which are specially configured to store and execute program commands. Examples of the program commands may include machine language code created by a compiler and high-level language code executable by a computer using an interpreter and the like. The hardware devices described above may be configured to operate as at least one software module to perform the operations of the disclosure, and vice versa.

[0128] In addition, the above-described method or apparatus may be implemented by combining all or part of constructions or functions thereof, or the constructions or functions may be separately implemented.

[0129] Although the present disclosure has been described above with reference to the embodiments of the present disclosure, those skilled in the art may variously modify and change the present disclosure without departing from the spirit and scope of the present disclosure as set

forth in the claims below.

DESCRIPTION OF SYMBOLS

[0130] **10**: projector [0131] **11**: power supply part [0132] **12**: light source generation part [0133] **13**: vehicle information receiving part [0134] **14**: display mirror monitoring part [0135] **15**: distortion correction part [0136] **20**: optical reflector [0137] **21**: transparent film layer [0138] **22**: reflective layer [0139] **23**: adhesive layer [0140] DM: display mirror

Claims

1. A vehicle display system, comprising: a projector configured to emit projection light toward an optical reflector; an optical reflector positioned in one region in a vehicle to reflect the projection light; and a display mirror configured to display a virtual image, which is visible to a user, in a vehicle as reflected light reflected on the optical reflector is incident, wherein, a polarizing filter having a polarization axis in a first direction is coated on an external lens of the projector, the display mirror is one region of a windshield, made of glass, a window, and a transparent flat plate, made of glass or acrylic, in the vehicle, and a polarizing filter having a polarization axis in a second direction perpendicular to the first direction is coated on an inside of the display mirror.
 2. The vehicle display system according to claim 1, wherein the projector is positioned in one region of an overhead console located at a top of a vehicle based on a driver, pillar A in the vehicle, pillar B in the vehicle, an armrest of a rear row of the vehicle, and a ceiling of the rear row of the vehicle, the optical reflector is positioned in one region of a dashboard and ceiling in the vehicle and an inside of a door, and the display mirror is one region of the windshield.
 3. The vehicle display system according to claim 1, wherein the optical reflector comprises a meta-surface on which a plurality of nanostructures are arranged and which diffracts and reflects the projection light according to a shape and arrangement form of each of the plurality of nanostructures, the plurality of nanostructures have a thickness thinner than a wavelength of the projection light, and an arrangement distance between the plurality of nanostructures is shorter than the wavelength of the projection light.
 4. The vehicle display system according to claim 3, wherein the optical reflector comprises: a substrate; and the plurality of nanostructures arranged on the substrate and configured to function as a meta-surface, wherein the substrate comprises: a transparent film layer configured to transmit the projection light; a reflective layer positioned under the transparent film layer; and an adhesive layer positioned under the reflective layer.
 5. A vehicle display system, comprising: a projector configured to emit projection light to a display mirror; the display mirror configured to reflect the projection light and transmit first reflected light to an optical reflector, and, as second reflected light reflected on the optical reflector is incident, to display a virtual image that is visible to a user in a vehicle; and the optical reflector positioned in one region in the vehicle to reflect the first reflected light to the display mirror, wherein, a polarizing filter having a polarization axis in a first direction is coated on an external lens of the projector, the display mirror is one region of a windshield, made of glass, a window, and a transparent flat plate, made of glass or acrylic, in the vehicle, and a polarizing filter having a polarization axis in a second direction perpendicular to the first direction is coated on an inside of the display mirror.
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